

Potentials

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1 Electrical Potentials

Electric potentials are measures of the potential energy created by an electric field. You can think of them as such:

Potential: Potential Energy :: Field: Force Energy

For the potential energy of point charges, there is a very big positional element here. As long as you have two charged particles, as long as they want to move somewhere, there will always be potential energy in that system. Usually, systems want to reduce that energy to zero.

1.1 Formulas and Dimensional Analysis

Let's look at the formulas for volts and field, and see what we can make for them. Let's have the formula for volts, and put of the formula for electric field for comparison.

$$V = \frac{U}{q}$$

$$\vec{E} = \frac{\vec{F}}{q}$$

Let's do a quick dimensional analysis too for these *volts*:

$$[V] = \frac{[J]}{[C]}$$

Differences between field and potential:

- Field is a vector quantity, voltage is a scalar quantity
- Field lines are always perpendicular across vector lines

Let's have a visualization changes of potential over time:

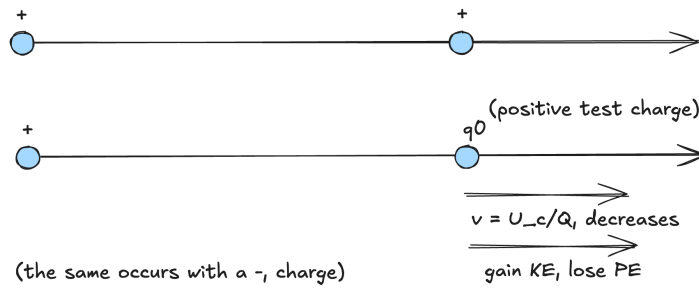


Figure 1: The change of charge across a number line

Notes:

- Voltage decreases as one goes through a field lines

1.2 Equipotentials

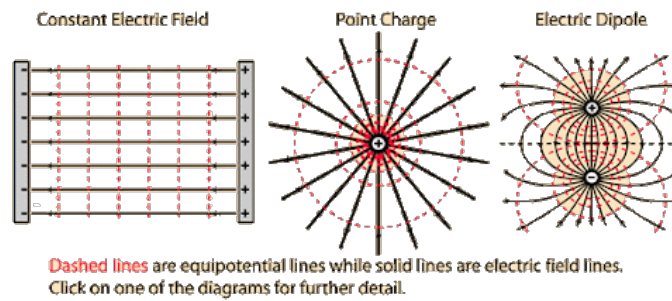


Figure 2: Example of electric fields vs electric potentials

- Lines across these show the areas of equidistant potentials
- Circular around point charge
- *Perpendicular to field lines*, (e.g. Electric field is the negative gradient of potential), or $E = -\Delta V$

1.3 Calculating Electrical Potential from Electric Field

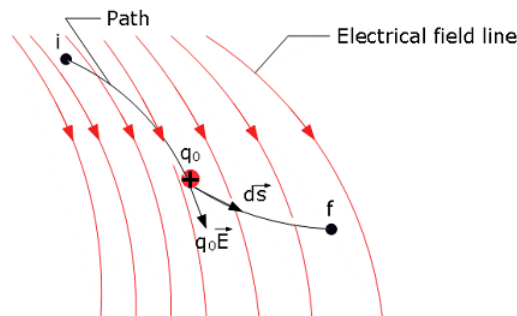


Figure 4-3 A test charge q_0 moves from point i to point f along the path shown in a nonuniform electric field. During a displacement $d\vec{s}$, an electric force $q_0\vec{E}$ acts on the test charge. This force points in the direction of the field line at the location of the test charge.

Figure 3: Visualization of the different differentials here

To try to find a formula for this, starting from the givens and working up.

$$\vec{F} = q_0 \vec{E}$$

$$W = \int \delta W$$

$$\begin{aligned} W &= \int \delta w = \int_i^f \vec{F} \cdot \delta \vec{s} \\ &= -\Delta U = -q_0 \Delta V \\ \Delta v &= \boxed{- \int_i^f \vec{E} \cdot \delta \vec{s}} \end{aligned}$$

Given that electric field is a vector, and potentials are scalars, one might have to treat these as partial derivatives of x and y. So we have the original equation:

$$\vec{E} = \frac{Q}{4\pi\epsilon R^2} = -\frac{\delta V}{\delta s}$$

But we have to split it into components:

$$E_x = \frac{\delta V}{\delta x}$$

$$E_y = \frac{\delta V}{\delta y}$$

$$\vec{E} = -\frac{\Delta V}{\Delta S} = \frac{\delta V}{\delta x} + \frac{\delta V}{\delta y}$$

2 The Electric Potential of a Point Charge

Given a ring of radius R, which has a uniformly distributed charge Q, we can infer the following:

- The net electric field in the center is zero (common sense)
- The net electric field at every point is also zero (similar to gravitational field)
- What is the potential inside the ring, given $\Delta V = \int \vec{E} \cdot \delta \vec{s} = \int 0 \cdot \delta \vec{s} = \boxed{C}$.

2.1 Calculating the Point Charge's Potential Through Integration

Things to note:

- Potential is a function of charge and radius, or the distance from a point
- Charge creates a scalar potential field
- $u(q_1, q_0) = q_0 V(q_1)$, the relationship between charge and potential energy

To get definite zero, we can set the potential at $r \rightarrow \infty$ to be zero, or $\lim_{r \rightarrow \infty} V(q) = 0$ to be zero.

$$\begin{aligned} \Delta V &= V_f - V_i = 0 - V_i = \int_r^\infty \vec{E} \cdot \delta \vec{s} \\ -V_r &= - \int_r^\infty \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2} [\hat{r} \cdot \delta \vec{s}] = - \int_r^\infty \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2} \delta r \\ &\boxed{V_r = \frac{Q}{4\pi\epsilon r}} \end{aligned}$$

Note that the property is *scalar*. This makes stuff like addition and subtraction better!

3 Electric Flux

A measure of how many field lines pierces a certain surface / goes through a surface

3.1 Finding formula for Electric Flux

Flux is proportional to

- Cross sectional area of the surface [A]
- Strength of the wires [E]
- orientation of the wires [θ]

Max flux occurs when $\theta = 0^\circ$ (the angle between the normal of the surface and the field lines), and min flux occurs when $\theta = 90^\circ$ (e.g. when normal of the surface and the field lines are perpendicular).

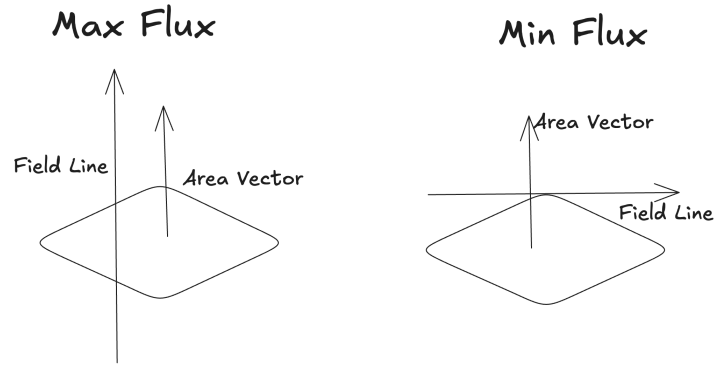


Figure 4: Visualization of max and min flux

Thus, you can compute the flux here:

$$\Phi = \vec{E} \cdot \vec{A} = |\vec{E}| |\vec{A}| \cos \theta$$

This is with the assumption that that

- Plane is regular
- Field is constant

For an irregular plane / field, you can use this formula

$$\Phi = \int \vec{E} \cdot d\vec{A}$$

3.2 Closed Areas vs. Open Areas

Gaussian Surfaces are closed figures, and that allows you to find a direction for the area vector, which is always outwards. Some additional points about closed areas and flux:

- The flux for an object that is hit by continuous equipotentials and doesn't contain any of their own is zero
- For example, if you have flux that goes through a box, the flux is zero

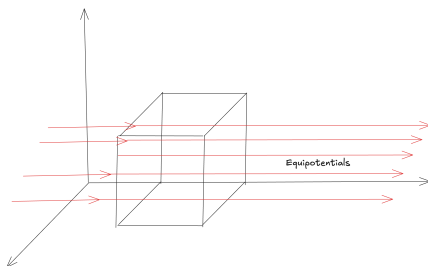


Figure 5: Object with zero flux

4 Gauss's Law

Electric flux gives us **Gauss' Law**, which tells us that the electric flux through a closed surface is proportional to the enclosed charge.

4.1 Gauss' Theorem

Gauss' theorem relates flux to charge, field, and area:

$$\Phi = \frac{q_{\text{enc}}}{\epsilon_0} = \int \vec{E} \cdot \delta \vec{A}$$

This can also be related to potentials, through the calc idea of differentiation (iirc)

$$\Delta \cdot E = \frac{\rho}{\epsilon_0}$$

These equations, though, are only good/useful if you have a good gaussian surface

4.2 Nice Gaussian Surfaces

An example is if it is symmetrical to a charged object. An example is a circular surface with a point charge in the center.

$$\begin{aligned}\Phi &= \frac{q_{\text{enc}}}{\epsilon_0} = \int \vec{E} \cdot \delta \vec{A} \\ &= E \int \delta A = \frac{Q}{\epsilon_0} \\ E(4\pi r^2) &= \frac{Q}{\epsilon_0} \\ E &= \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2}\end{aligned}$$

(Interesting because this merges both empirical physics with pure mathematics)

4.3 Electrical Fields vs Flux

Electric fields and flux can vary, for but flux you can calculate a total since it's a scalar value